

REAL ESTATE FIRM

KING COUNTY,
SEATTLE





GVL Pvt. Ltd.



Meet Our Team

At GVL Realty Group, we combine local expertise with a personalised approach to help you find your way home. Our boutique firm is built on a foundation of trust, market survey, and a commitment to seamless transactions.

Whether you are buying your first home or expanding your portfolio, WE are here to turn your real estate goals into reality.

Our team brings a diverse range of strengths to every deal:

- Gregori: Data Auditor
- Vitor: Pricing Specialist
- Laxmi: Market Navigator



GREGORI



VITOR



LAXMI

Our Company GOAL

- Preparing the variables is not a step before hiring the employees (model) — it is the EMPLOYEE (model)
- The quality of a prediction depends on how closely the variables resemble the reality they try to predict.
- Buyers are the ultimate decision maker only to the features of the listing you showcase.



Three things that stand out :

Before any hiring (modelling) decision, the dataset shape already signals what to expect.

\$7.7M

MAX_PRICE

Range from \$75K to \$7.7M.
Long right tail → skewed
distribution. Linear models
will struggle here.

33

MAX_BEDROOMS

Physically implausible for a
single-family home.
A clear outlier to clean.

1.65M

MAX_SQFT_LOT

1.65M square feet of land =
rural property, not urban
housing. Another long tail to
handle.

CLEANING WITH INTENT

Decisions reflect the market I want to model — not arbitrary deletions.



MISSING VALUES

No imputation strictly required, but the pipeline still imputes by median as a safety net for production data.



REPEATED IDS — KEPT

Same house, sold twice within the period. Each sale reflects different market conditions. Legitimate signal, not noise.



OUTLIERS REMOVED

Bedrooms 1-10, bathrooms 1-6, price within the 1st-99th percentile. The goal is the typical market, not luxury extremes.

ADDITIONAL VARIABLES

To make your life easy during the decision making, we narrow-down the multiple variables to straight-forward ones.

sale_year + sale_month

Macro cycle (2014 vs 2015) and seasonality (spring peak, school calendar).

house_age

Buyers value effective age, not absolute year. Captures wear and outdated systems.

was_renovated

Binary flag that separates two distinct populations: invested vs original condition.

years_since_renovation

A 2-year-old renovation does not equal a 30-year-old one.

has_basement

In the Pacific Northwest, basement presence is a real differentiator.

basement_ratio

Relative weight matters more than absolute basement size.

living_lot_ratio

Encodes building density — townhouse vs suburban with garden.

total_rooms

Functional summary of usable space — helps linear baselines.

THE VARIABLES WE DROPPED DELIBERATELY

The complicated additional choices which might affect the decision making for a buyer were deliberately removed.

ID

Identifier only. No predictive signal — would risk memorization in tree models.

date

Information already extracted into sale_year and sale_month.

sqft_above

By construction:
sqft_above +
sqft_basement =
sqft_living. Perfect multicollinearity.

sqft_basement

Same multicollinearity — and basement signal is already captured by has_basement and basement_ratio.

Removing redundant features is itself a modeling decision — it tells the model what NOT to learn.



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TOP CORRELATIONS of features with the price

The data confirms the real-estate theory: size, quality, and neighborhood lead.

READING THE CHART

sqft_living (0.70)

Price-per-square-foot is what appraisers use.

grade (0.67)

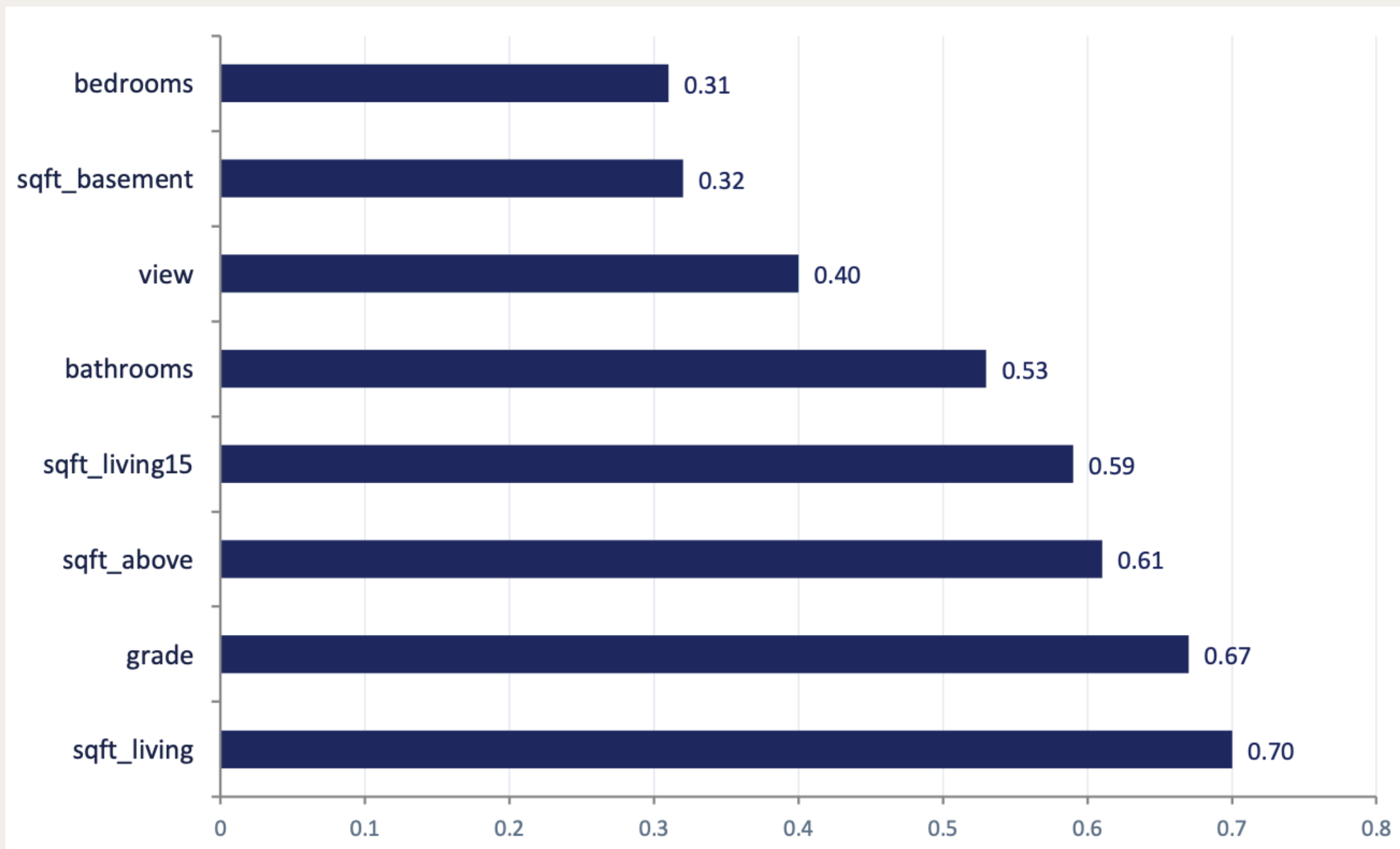
Construction quality — materials, finish, design.

sqft_living15 (0.59)

Average size of the 15 nearest homes — encodes the neighborhood.

Note

zipcode appears with near-zero correlation. Geography is not linear — it requires categorical treatment.



ZIPCODE - a label not just a number

Zipcode 98199 is not "greater than" 98001 in any economic sense.

BEFORE:

```
zipcode
98001 98002 98003 ...
```

Treated as int → spurious order:

"higher zipcode = higher price" ✗

AFTER:

zip_98001	zip_98002	zip_98003 ...
1	0	0
0	1	0
0	0	1

≈ 70 binary columns — no order assumed

PIPELINE SUMMARY

NUMERICAL

Median imputation → StandardScaler

CATEGORICAL

Mode imputation → OneHotEncoder (handle_unknown = ignore)

The Employee (model) does not learn from the raw dataset – it learns from how they understood the market needs and translated it into the dataset.

Algorithms are interchangeable. Variables are not.

Meet our Market Navigator & Builder



Defining the Target

Separating Signal from Noise

Our analysis begins by isolating the objective. We split our data into a feature matrix (X) and a target vector (y).

Target(y): The "Price" column—the ultimate value we seek to predict.

Features(X): 25 attributes dropped of administrative noise (IDs and Dates).

Data Integrity: Removed columns like sqft_above and sqft_basement to focus on independent primary features.

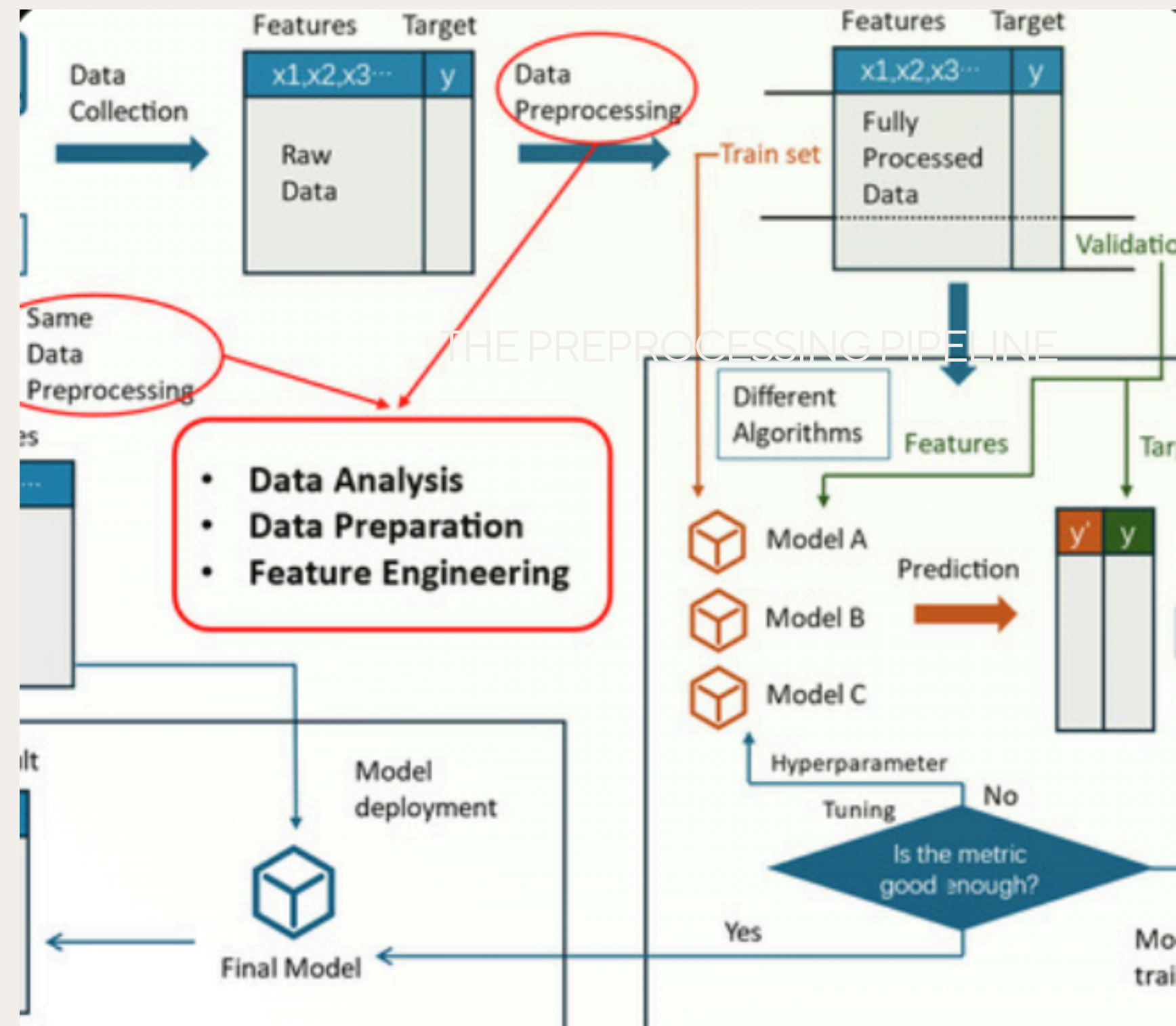




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TRAINING

We implement a strict 80/20 train-test split to ensure our real estate agent partners can trust the predictions on new listings





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The Split Validation

BUILDING A RESILIENT MODEL FOR OUR BUYERS

- Training Pool: 16,869 sales records used to learn market nuances.
- **The 20% Holdout:** 4,218 records sequestered to simulate "unseen" future sales.
- **Random State 42:** Ensured reproducibility and consistency across all testing phases.



NEXT.....

Let's meet our efficient employees:

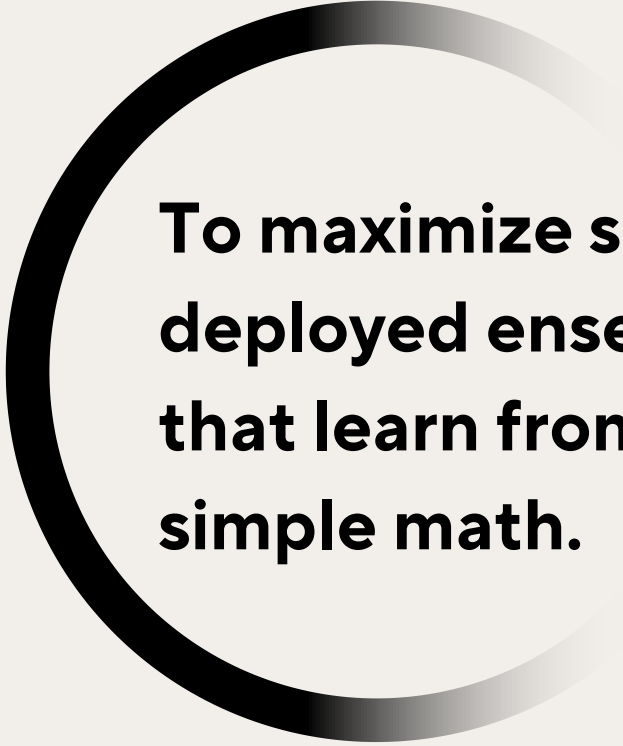
Setting the Benchmark for our employees

Baseline employees (models) establish the floor for our predictive performance.

Model Archetype	R-Squared R2	Mean Absolute Error (MAE)
Linear Regression	0.8421	\$78,278
KNN Regressor	0.7801	\$83,059

The Power of Employee Ensembles

Improved Model Architectures



To maximize sales accuracy, we deployed ensemble methods that learn from the errors of simple math.

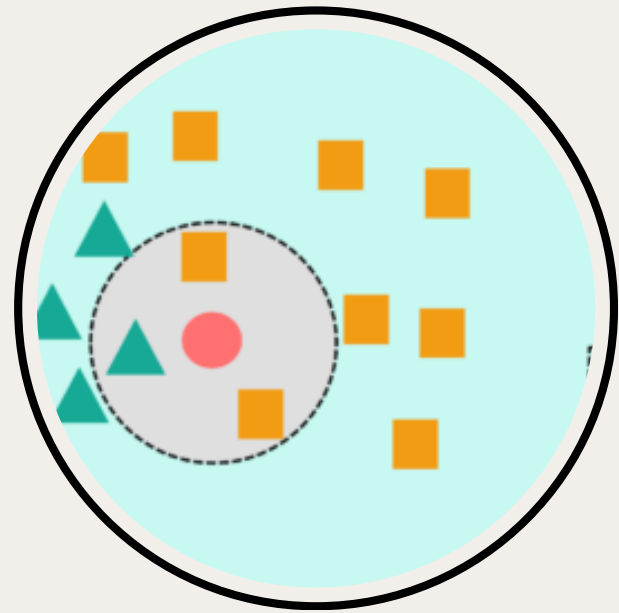
We have moved from 21,613 raw rows to a predictive engine that explains 86.6% of the market price volatility. Our validation strategy ensures your clients' investments are protected by rigorous data science.

- **Random Forest:** Utilizing hundreds of trees to capture non-linear market behaviors. Reached R2 of 0.866.
- **Gradient Boosting:** Sequentially refining predictions. Achieved R2 of 0.858.
- **The Edge:** Tuned ensembles reduced our average error by over \$15,000 compared to baseline KNN.

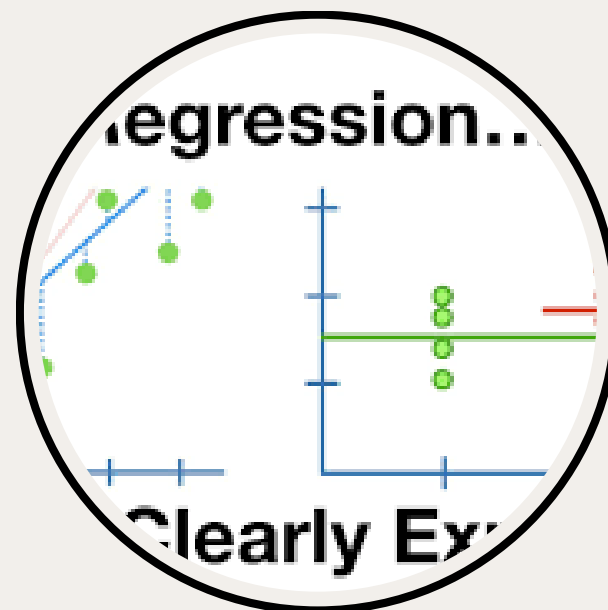


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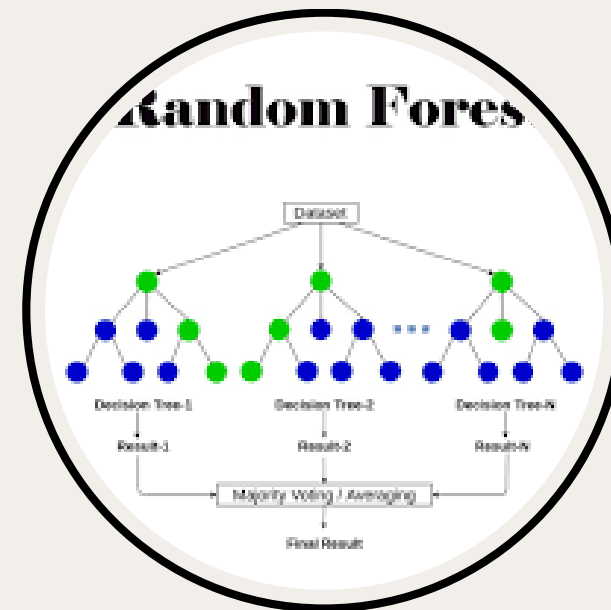
Comparing the EMPLOYEES



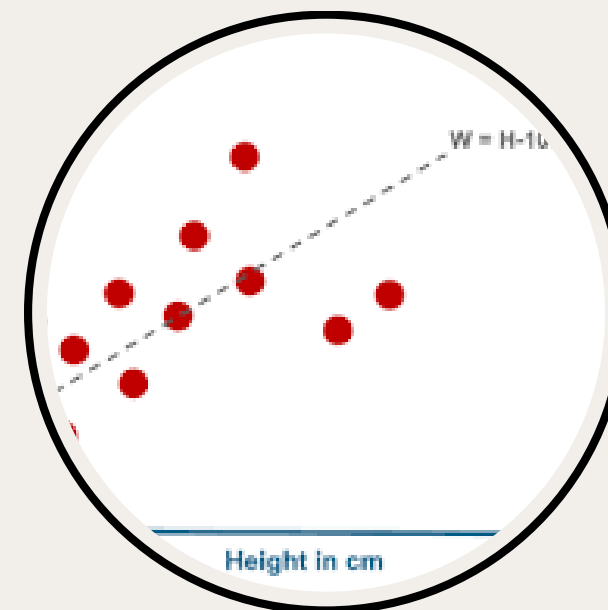
KNN



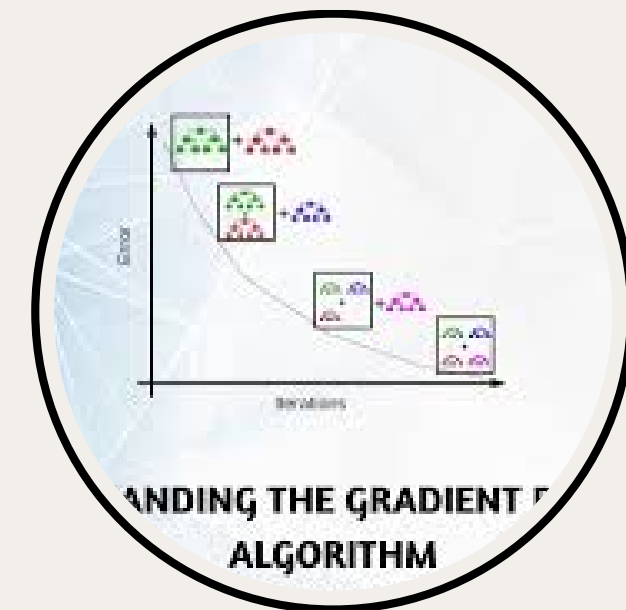
Ridge



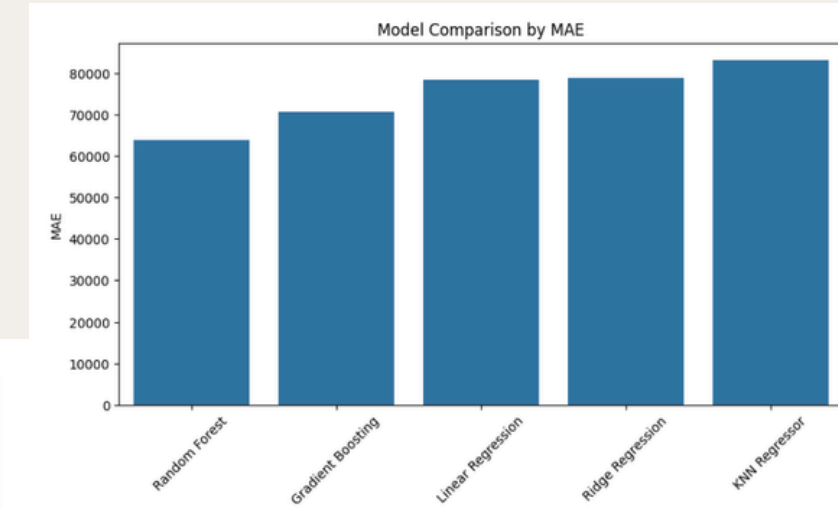
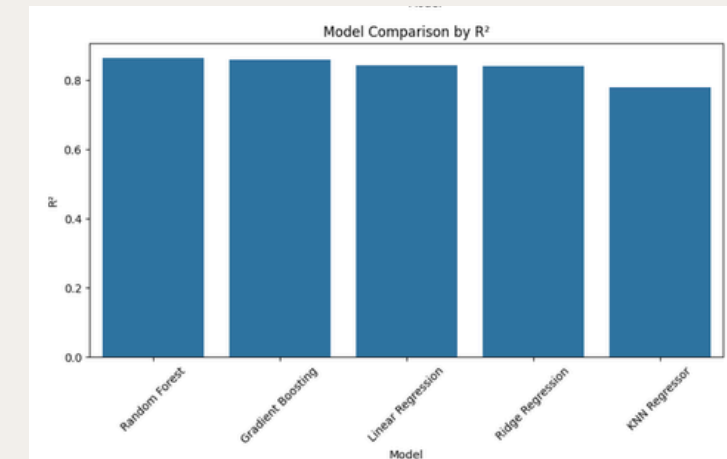
Random Forest



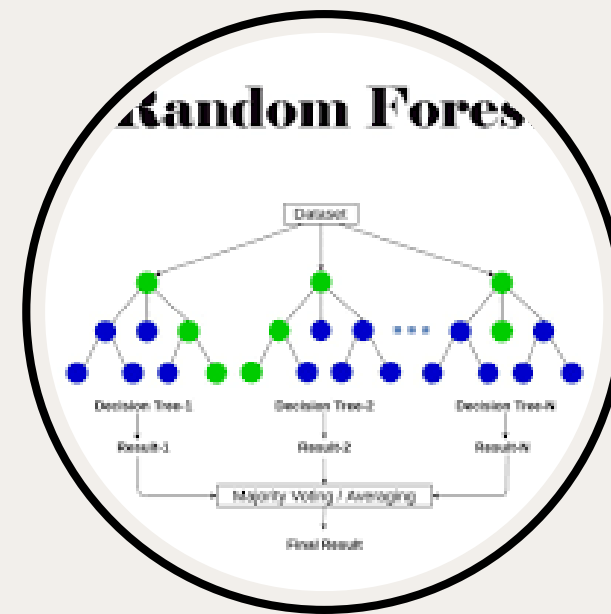
Linear Regression



Gradient Boosting



EMPLOYEE OF THE QUARTER



Random Forest

- Key metrics:
- Highest R2 (~0.864)
- Lowest MAE (~\$63.9K)
- Ensemble models outperform linear models
- Random Forest employee captures non-linear relationships better
- Simpler employee show slightly weaker performance No sign of underfitting, which means employees already perform reasonably well
- Slight caution: More complex employee (models) could have potential overfitting risk.

LETS NOT HIRE ANYBODY ANYMORE!

Hyper parameter Tuning

Or lets call, making the employee of the year work more...!

Tuning the employe of the year a bit more to find out better performance probability.

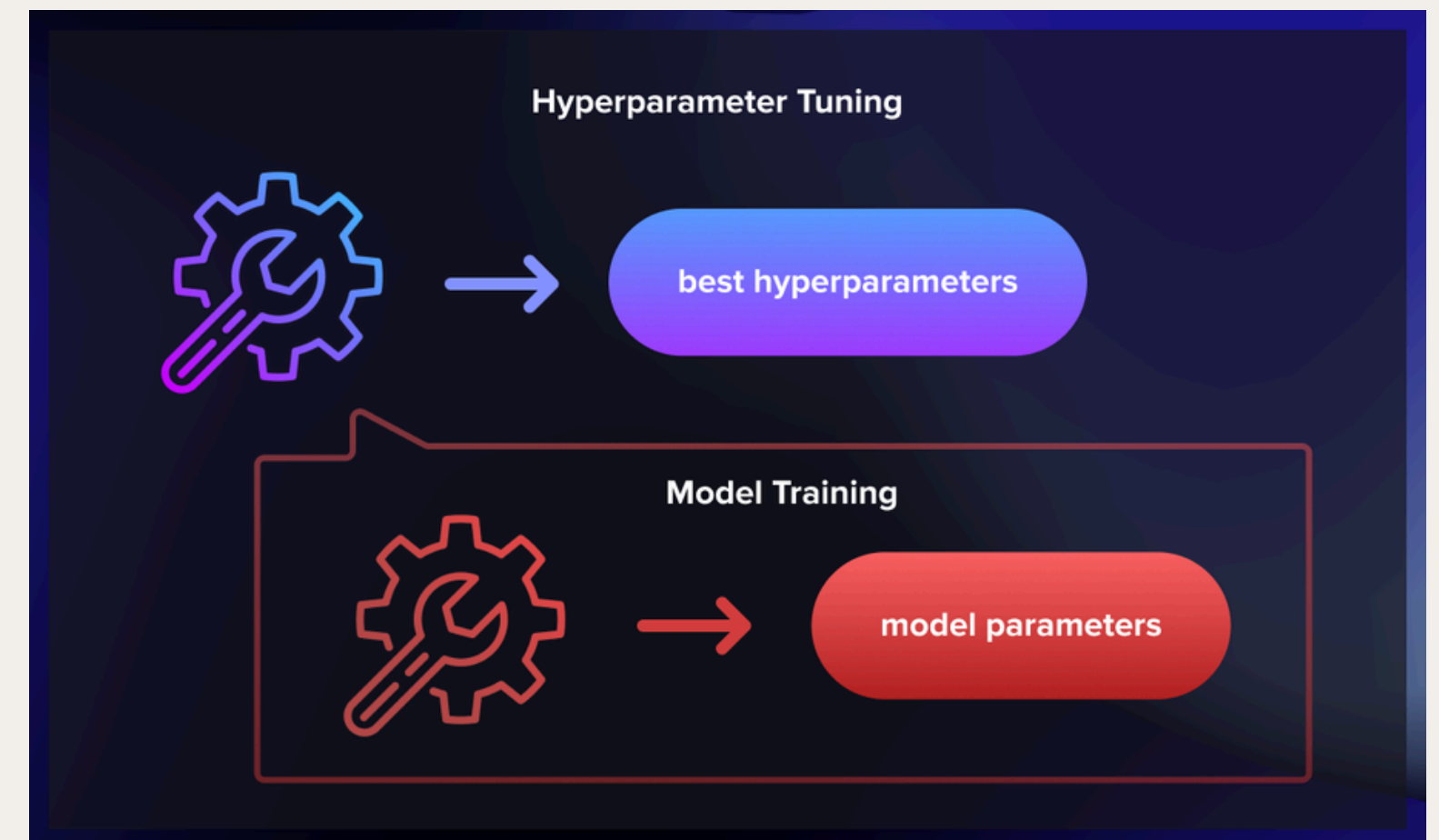
We added a new skill to be trained for our employee,
RandomizedSearchCV

Goal: reduce MAE and improve generalisation

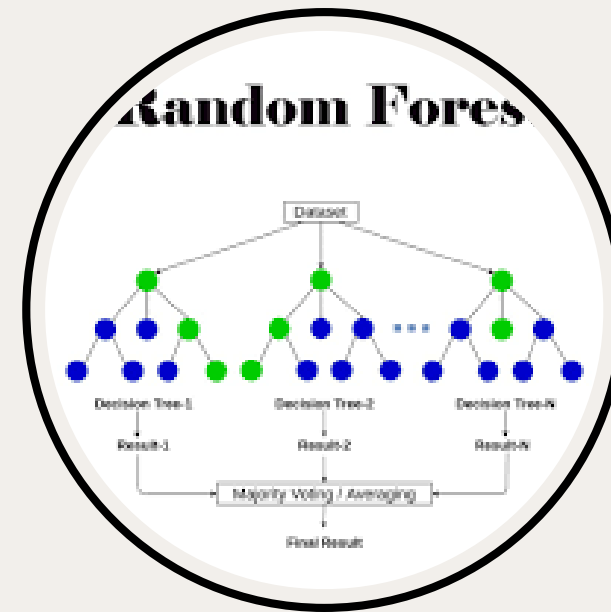
- **Result:**
- **MAE decreased to ~\$63.2K**
- **R2 increased to ~0.867**

Random Forest has highest overfitting risk

- **Tuning:**
- **limits tree depth**
- **controls splits**
- **reduces variance**
- **Improves generalisation, not just performance**



EMPLOYEE OF THE QUARTER



Tuned Random Forest

- Metrics:
- $R^2 \approx 0.867$
- $MAE \approx \$63K$
- $RMSE \approx \$103K$

- **Key Insights:**

Best balance between:

-accuracy

-robustness

-MAE used as main decision metric (given its real-world meaning)

- **Observations:**

After tuning, the Random Forest employee (model) remained the best performer. It was selected based on MAE, since it directly reflects the average error in dollars. On average, the model is off by about \$63,000, which is the lowest among all models tested.

Our Final Employee Training:

- Final model is tuned Random Forest
- Re-trained on training data
- Evaluated again on test set
- Results remain consistent

No change in metrics - as expected

Confirms:

- stability
- reproducibility

- The final employee (model) was then retrained using the best parameters followed by re-evaluation.
- The results remained the same, which confirms that the employee (model) is stable and reproducible.
- This step doesn't improve performance, but it formalises the final model that will be used.

Coming back to our predictions and reality of the PRICES

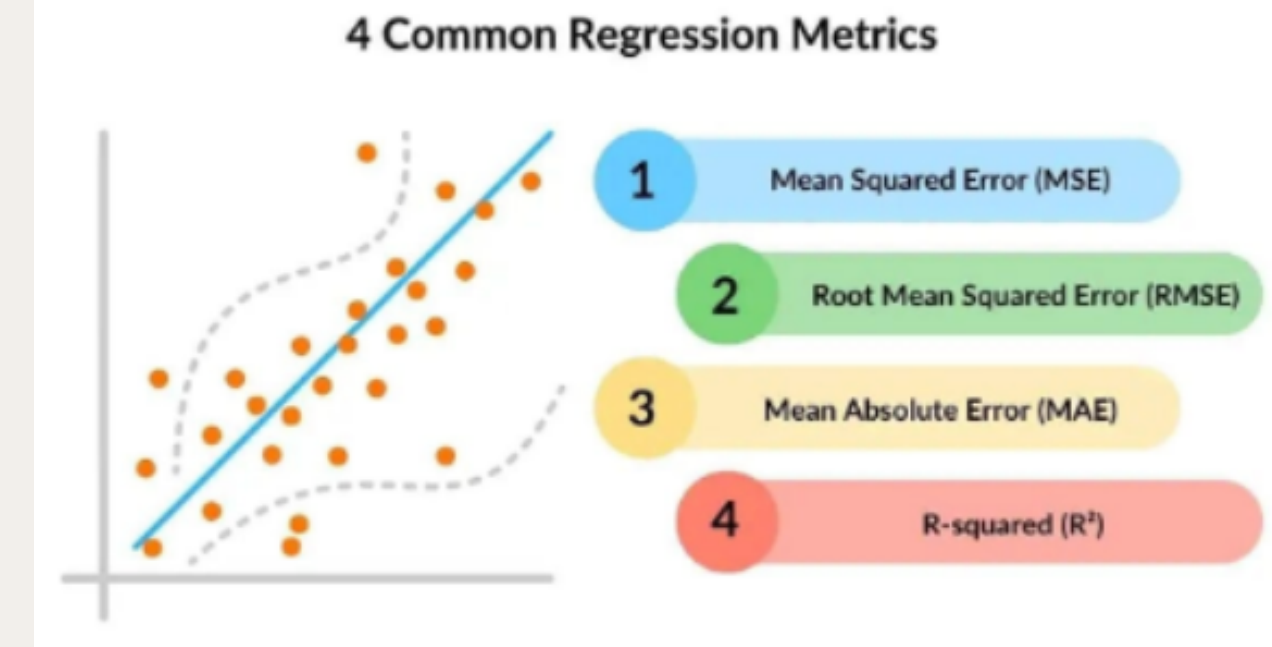
Scatter plot: diagonal hints into perfect predictions

- Predictions follow the general trend
- Errors increase for expensive houses
- Model captures overall structure well
- A comparison between real house prices and predicted values was made.
- Most predictions follow the diagonal line, which represents perfect predictions.
- This shows that the model captures the overall pricing pattern well.
- However, we observe larger deviations for high-value properties, which are inherently more difficult to predict.



Measuring Predictive Success

We judge our "Budget-Friendly" engines based on three industry-standard metrics:



Mean Absolute Error (MAE):
Average dollar-amount off. In our best model, this was reduced to ~\$63,188.

Root Mean Squared Error (RMSE): Penalizes large errors, ensuring investment safety.

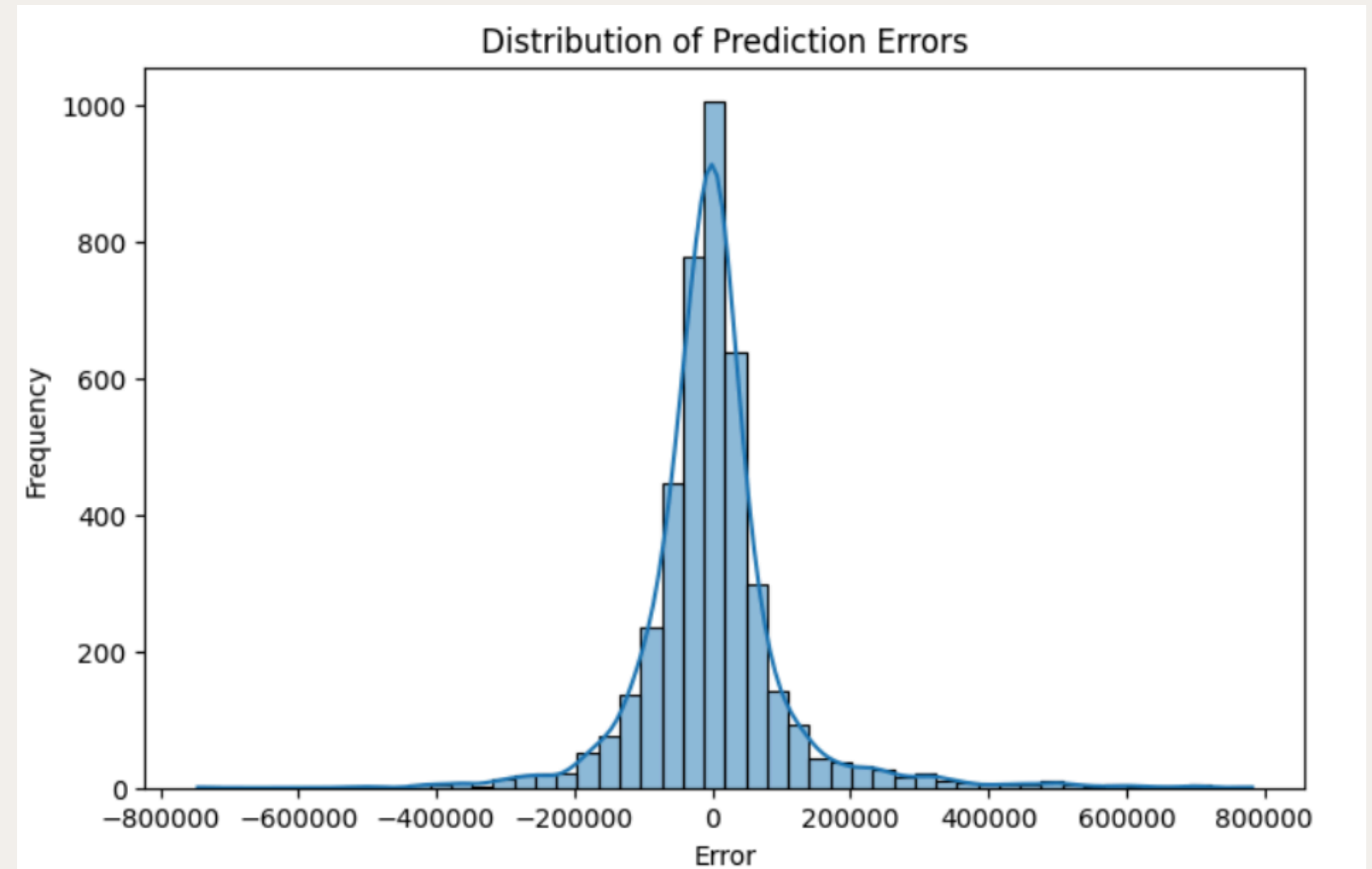
R-Squared R^2 : The percentage of variance explained by our logic.

ERROR ANALYSIS

- **MAE $\approx$ \$63K**
- **Median error $\approx$ \$36K**
- **Distribution of errors (histogram)**

Median < Mean, meaning that most predictions are closer than MAE suggests

- Large errors are rare but impactful
- No strong overfitting:
- model performs well on test data
- Looking deeper into the errors, one notices that the median error is around \$36,000, which is significantly lower than the average error.
- This means that most predictions are relatively accurate, while a small number of large errors increase the average.
- Importantly, strong performance on the test set suggests there is no clear evidence of overfitting.

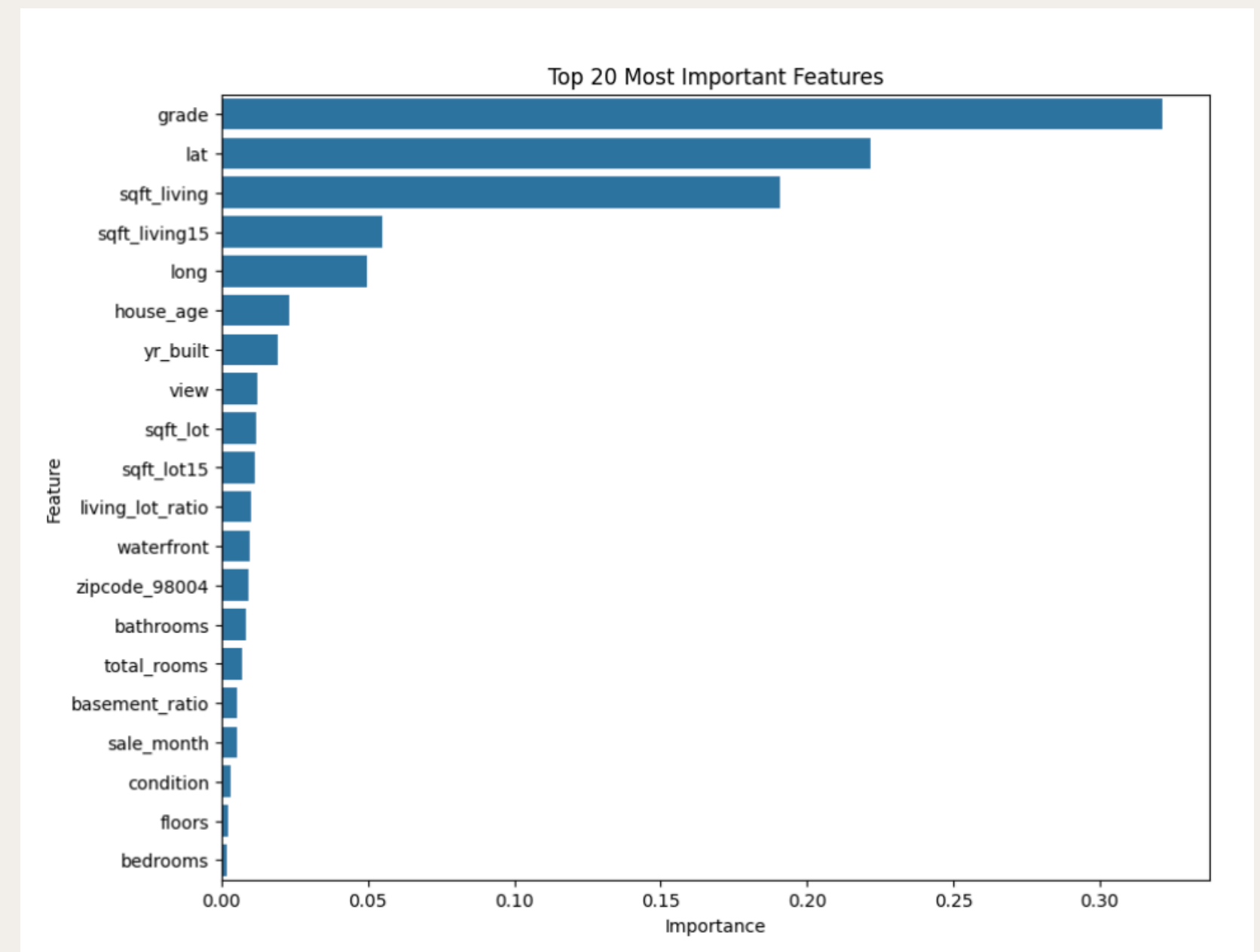


IMPORTANCE OF FEATURES

- grade
- location (latitude, longitude)
- sqft_living
- neighbourhood features

Price driven by:

- quality
 - location
 - size
 - matches real-world intuition
-
- Feature importance from the Random Forest model was then analysed.
 - The most influential features are grade, location, and living space.
 - This aligns with real-world real estate logic, where property quality, location, and size are the main drivers of price.



FINAL CONCLUSION

- **Best model: Tuned Random Forest**

Key drivers:

Quality

Location

Size

No strong underfitting

No strong overfitting

Tuning improved generalisation

Models are:

Well-fitted

Stable

Random Forest complexity managed correctly

TO CONCLUDE:

- The Tuned Random Forest model provided the best performance and generalisation.
- There is no evidence of strong underfitting, as all models achieved reasonable results.
- There is also no clear evidence of overfitting, since the final model performs well on unseen data.
- However, due to the complexity of Random Forest, hyperparameter tuning was essential to control overfitting risk and ensure reliable predictions.
- Overall, the model confirms that house prices are mainly driven by property quality, location, and living space.





THANK YOU



for choosing GVL !

